Vedika Sunil Bang

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Education

Georgia Institute of Technology **Aug 2022 – Dec 2023**
MS in Cybersecurity (GPA: 3.63/4.0)(Highest Honors) *Atlanta, GA*

- Relevant Coursework: Information Security Policies, Applied Cryptography, Intro to Malware Reverse Engineering, Advanced Network Security, Secure Computing Systems, Advanced Operating Systems, Defense Lab, Cyber Practicum

Graduate Teaching Assistant – CS 6725 Aug 2023 – Dec 2023 Introduction to Information Security Policies

Pune Institute of Computer Technology **Aug 2016 – Nov 2020**
BE Electronic and Telecommunications (First Class with Distinction) *Pune, India*

Experience

Georgia Tech **March 2024 – Present**
Cybersecurity Research Associate *Atlanta, GA*

- Analyzing password-stealing trojans against open-source password managers, focusing on MITRE ATT&CK's TA0006 Credential Access to identify vulnerabilities and improve detection and prevention strategies

Stroz Friedberg, an Aon company **Jun 2023 – Aug 2023**
Cyber Summer Associate *Washington, DC*

- Led the identification and documentation of tactics, techniques, and procedures (TTPs) by analyzing MSP tools such as RMMs - ScreenConnect, and AnyDesk, integrating over 40 unique Indicators of Compromise (IoCs) into Aon's SIEM , enhancing threat intelligence capabilities and incident response efficiency by 35%
- Performed comprehensive host-based analysis on over 70 digital forensic images using tools like X-ways, EnCase, and FTKImager, while performing log analysis on firewall logs, VPN logs, network traffic, O365 and AWS, Azure environments' logs for a Business Email Compromise (BEC) to identify potential Adversarial TTPs
- Developed Python and PowerShell scripts to automate routine tasks with custom scripts, reducing manual workload by 10%, while assisting on a Incident response engagement

Information Sharing and Analysis Center **Dec 2021 – May 2022**
Computer Forensics Engineer *Delhi, India*

- Spearheaded comprehensive investigations into live systems utilizing Volatility3 for memory analysis and LiME for memory extraction. Leveraged IDA-Pro, Olly-DBG, & Radare2 for static analysis
- Developed and successfully wrote 5+ critical exploits for Windows as part of the OpenCTI team to evaluate, test, and integrate new security controls
- Dockerized, and tested 200+ specialized labs for ISAC's virtual Capture The Flag (CTF) event, covering a diverse range of OWASP Top 10 challenges

Wattlecorp Cybersecurity Labs **Aug 2021 – Dec 2021**
Vulnerability and Penetration testing Intern *Remote, India*

- Performed vulnerability assessments, penetration tests, and code reviews on 6 assigned systems, utilizing both SAST (Static Application Security Testing) and DAST (Dynamic Application Security Testing) methodologies to identify and mitigate security risks - reported and documented DOM-XSS

Projects

Living off The Land Attacks: Application IoC Generator **Aug 2023 - Present**

- Developed a python-based tool for automating the generation of high-fidelity IoCs and detection rules, achieving a 60% acceptance rate for enhancements. This tool streamlined the integration with SIEM/EDR, prioritizing actionable, high-fidelity alerts and significantly improving incident response efficiency

Mini Coursework Projects **Jan 2023 - Dec 2023**

- Development of Host based and Network based IDS: CS 6264 Defense labs
- Implemented PoC for exploits involving XSS, CSRF, SQL Injection: CS 6035 Intro to Information Security

Sending Email from a Residential ISP, CS 8803-EMS **Sep 2022 - Dec 2022**

- Led an experimental study with **Prof. Paul Pearce** to explore the individual impacts of removing email security protocols (SPF, DKIM, DMARC) on the deliverability and security of emails sent from residential IP addresses; providing insights into email server performance and reliability in home internet setups

Malware HomeLab *Tech stack: IDA Pro, Ghidra, Olly-dbg, C, x86_64/32* **Jan 2023 - Present**

- Engaged in solving complex CTF challenges on platforms like Root Me, CrackMes.one, and OverTheWire. Gained practical experience in reverse engineering malware such as Michelangelo.1, DOS-7, Harulf, SQLSlammer, Lucius by manual and dynamic code review; Participated in NSA Codebreaker Challenge ('23), contributing to Georgia Tech's #1 winning team

Technical Skills

Operating System: GNU/Linux, Kali (Pentesting), Windows (Malware Analysis)
Languages: Python, C, C++, SQL, Bash, PowerShell, Javascript
Security Tools/Protocols: Volatility3, Wireshark, IDA Pro, Ghidra, GDB, Angr, Splunk, Xways, FTKImager, SQLite, Nmap, Metasploit, Burp Suite, YARA, Semgrep playground, OpenSSL, Snort3, Suricata, Autopsy
Frameworks/others: Git, Jenkins, Docker, SIFT, MITRE ATT&CK, AWS, ISO 27001, PCAP analysis

Certifications

HackTheBox Certified Pentester HackTheBox
Currently pursuing OSCP
Google Cybersecurity Professional Certificate Google (Coursera)